

Astra AI Wellness Companion

End-to-End Secure UI/UX Flow Architecture

This document outlines the complete patient journey from the opening screen to checkout, detailing how the newly implemented security and anti-hijacking layers integrate into the user interface.

Screen 1: Splash & Secure Authentication

Visuals: AyurEze branding fades in. A loading spinner indicates server connection.

- **User Action:** Opens the app.
- **System Action:** The app verifies the Firebase JWT token.

■ *Security Layer: If the SSL certificate is within 14 days of expiration, the background Doomsday Monitor has already alerted DevOps, ensuring this screen never hangs.*

Screen 2: The Main Chat Interface

Visuals: A clean, continuous chat feed. A text input box and a microphone button sit at the bottom.

- **User Action:** Types 'What herbs help with digestion?' and hits send.
- **System Action:** The Flutter app uses ListView.builder to render the chat bubbles without crashing the phone's memory (Fix #29).

■ *Security Layer (Token Budget): Before the AI replies, Redis checks if the user exceeded 2,000 tokens/hr. If safe, the AI generates a response.*

■ *Security Layer (XSS Sanitizer): The AI's response is scrubbed of malicious HTML/JS before appearing on the screen.*

Screen 3: Voice Confirmation Gate

Visuals: A modal popup card temporarily blocks the chat interface.

- **User Action:** Taps the microphone and says 'I took fifty milligrams.'
- **System Action:** The Flutter app pauses the flow and presents the text to the user.
- **UI Element:** 'I heard: I took 50 milligrams. Is this correct? [Confirm] / [Edit]'

■ *Security Layer (Voice Triage): Prevents the AI from acting on dangerous transcription errors caused by background noise (Fix #21).*

Screen 4: Abuse Prevention (Account Frozen)

Visuals: The standard chat keyboard disappears. The screen flashes a red warning banner.

- **User Action:** A hacker writes a bot to spam the chat box rapidly.
 - **System Action:** The backend returns metadata={'account_frozen': True}.
 - **UI Element:** '■■■ ACCOUNT FROZEN FOR REVIEW. Unusually high activity detected.' The chat input is permanently disabled.
- *Security Layer (Denial of Wallet): Protects company API credits from being drained (Fix #31).*

Screen 5: Safety Lock & SOS Escalation

Visuals: The AI stops responding. A prominent red 'SOS Emergency' button overrides the chat box.

- **User Action:** Patient types 'My chest is crushing me.'
 - **System Action:** The AI checks the Cultural Protocol. If verified as a real emergency, it locks the chat.
 - **UI Element:** '■ CHAT LOCKED. Critical emergency detected. Tap SOS to call 112.'
- *Security Layer (Medical Liability): The exact prompt and AI lock response are permanently etched into the Immutable Audit Ledger in Supabase (Fix #12).*

Screen 6: Smart Cart & Doctor Handoff

Visuals: An interactive product carousel appears inside the chat feed containing prescribed medicines.

- **System Action:** The Doctor clicks 'Generate Prescription' on the admin panel.
 - **System Action:** A massive Shopify JSON payload is sent without hitting the 1MB Nginx limit (Fix #42).
 - **UI Element:** Patient sees 'Dr. AyurEze has prescribed: Ashwagandha. [Add to Cart]'
- *Security Layer (Prompt Injection): The AI itself cannot generate this cart. It requires a Doctor's authorized JWT token.*

Screen 7: Prescription PDF & Forgery Scanner

Visuals: A built-in PDF viewer showing the official prescription document with an embedded QR code.

- **System Action:** The background Celery Worker successfully generated the PDF without crashing the main server (Fix #39).
- **User Action:** The patient downloads the PDF and takes it to a pharmacy.
- **UI Element (Pharmacy Portal):** A 'Scan Prescription QR' button opens the camera.

■ *Security Layer (Non-Repudiation): Scanning the QR hits the /verify endpoint to cryptographically prove the patient didn't photoshop the medicine quantity (Fix #28).*

Screen 8: Checkout & Background Autopilot

Visuals: Standard Razorpay checkout UI, followed by a 'Success' checkmark.

- **User Action:** Patient pays for the medicine.
- **System Action:** The order clears in Shopify.
- **System Action:** The next morning, the background Cron job wakes up and sends a WhatsApp message: 'Don't forget to take your Ashwagandha!'

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